VTU-ETR Seat No.

Question Paper Version : A

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) May – 2024

Computer Science and Engineering / Information Science and Engineering

Time: 3 hrs. Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
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PART – I

(Research Methodology)

[50 Marks]

1. Which of the following research is also known as collaborative research?
a) Analytical research
b) Applied research
c) Action research
d) Descriptive research

2. Cross-sectional surveys are divided into two types, which are?
a) Exploratory and causal
b) Functional and Dysfunctional
c) Relational and logical
d) Descriptive and analytical

3. What are the main purposes of data analysis?
a) Description
b) Construction of Measurement Scale
c) Generating empirical relationships
d) Explanation and prediction

4. __________ involve a set of predetermined questions and highly standardized techniques of recording?
a) Structured interview
b) Unstructured interview
c) Interview guided
d) All of these

5. Which of the following is a primary step in social science research?
a) Preparing the Research Design
b) Developing the Research Hypothesis
c) Formulation of research problem
d) Execution of the Project

6. Find an example of probability sampling.
a) Convenience or accidental sampling
b) Purposive or judgmental sampling
c) Quota sampling
d) Stratified random sampling

7. Which of the following is not a data collection method?
a) Observation
b) Schedules
c) Interview
d) Research question

8. The term 'ethno' refers to
a) Geographical area
b) Social life
c) Cultural group
d) People or culture

9. Ogives is also known as?
a) Cumulative frequency curves
b) Frequency distributions
c) Transcendental Curve
d) All of these

10. Which of the following strategies is used to determine how common, widespread, or pervasive a certain trait, phenomenon, attitude, or opinion is in a demographic population?
a) Cross-sectional descriptive survey
b) Factorial survey method
c) Total survey design
d) Pooled surveys

11. Quantitative research involves conducting research using the following :
a) Feelings and hidden emotions, expectations
b) Facts, Numbers and numerical data
c) Humour and gratitude
d) Ethics, Accuracy and discipline

12. Which of the following best describes quantitative research?
a) the collection of non-numerical data
b) an attempt to confirm the researcher's hypotheses
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13. Which of the following includes examples of quantitative variables?
a) age, temperature, income, height
b) grade point average, anxiety level, reading performance
c) gender, religion, ethnic group
d) both a and b

14. Quantitative research only works if
a) You talk to the right number of people
b) You talk to the right type of people
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15. An image, perception or concept that is capable of measurement is called _______
a) Scale b) Hypothesis c) Type d) Variable

16. Why do you need to review the existing literature?
a) To make sure you have a long list of references
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a) Extend the statement of purpose by specifying exactly the question (the researcher will address)
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19. Concept is of two types
a) Abstract and coherent b) Concrete and coherent
c) Abstract and concrete d) None of these

20. _________ is concerned with discovering and testing certain variables with respect to their association or disassociation.
a) Exploratory b) Descriptive
c) Diagnostic d) Descriptive and diagnostic

21. In the process of conducting research 'Formulation of Hypothesis' is followed by
a) Statement of Objectives b) Analysis of Data
c) Selection of Research Tools d) Collection of Data

22. The data of research is _______

a) Qualitative only b) Quantitative only c) Neither of them d) Both a and b

23. _______- test is used to prove hypothesis of smaller sample.

a) t b) f c) z d) P

24. _____ tailed test is used when the researcher's interest is primarily on one side of the issue.

a) 1 b) 2 c) 3 d) 4

25. ______ is a part of the universe that can be used as respondents to survey.

a) Frame b) Sample c) Hypothesis d) Population

26. _______ Hypothesis states that there is no relationship between two or more variables

a) Null b) Alternative c) Negative Page d) Positive

27. _______ research helps to solve practical problems.

a) Qualitative b) Applied c) Basic d) Descriptive

28. The final stage in a survey is

a) Field work b) Assignment c) Calculation d) Reporting

29. ______ mean refers to the value obtained by dividing the sum of the values of all items by the total

a) number of items b) Arithmetic c) Geometric d) Harmonic

30. _______ is used to analyse differences between group means and the associated procedures

a) ANOVA b) Chi square c) Z-test d) correlation

31. Z test is used to test hypothesis when the sample size is ______

a) more than 30 b) less than 30 c) less than 50 d) none of these

32. The mode of 2, 5, 6, 4, 2, 7, 2, 5 is

a) 5 b) 6 c) 2 d) 7

33. Hypothesis should be ______

a) Any statement b) Empirically testable
c) Not empirically testable d) Can't say

34. What is the purpose of doing research?

a) To identify problem b) To find the solution
c) Both a and b d) None of these

35. Which ONE of these sampling methods is a probability method?

a) Quota b) Judgment
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36. _______ is the first step of the Research process

a) Formulation of a problem b) Collection of Data
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37. If the critical region is located equally in both sides of the sampling distribution of test statistics, the test is called
a) One-tailed b) Two-tailed c) Right tailed d) Left tailed

38. The standard deviation is always ______
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39. Mean, Median and Mode are
a) Measures of deviation
b) Measures of dispersion
c) Measures of control tendency
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40. A tentative proposition subject to test is
a) Variable b) Hypothesis c) Data d) Concept

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PART – II

Discipline Oriented Section

(Computer Science and Engineering / Information Science and Engineering)

[50 MARKS]

51. Consider a single dimensional array A[1..1000]. Base address 2000 and the size of each element is 4. Find the location of A[50]

a) 2186 b) 2126 c) 2096 d) 2196

52. Which data structure is used to remove recursion from a program?

a) Array b) Stack c) Queue d) List

53. Which of the following data structure is used to implement a priority queue in an efficient way?

a) Binary Heap b) Array c) Dequeue d) None of these

54. A linked list has O(1) time complexity for the following operation.

a) Insertion at the middle b) Deletion at the end
c) Insertion at the beginning d) None of these

55. What is the average time complexity of merge sort?

a) $\theta(\log n)$ b) $\theta(n \log n)$ c) $\theta(n^2)$ d) None of these

56. In mathematical logic clues a professor discuss the relationship between statements involving universal and existential quantifier consider following 2 statements:

1) "For every real number x, there exists a real number y such that $y = x^2$ ".
2) "There exists a real number y such that for every real number x, $y = x^2$ ".

Which of the following best evaluate the truthfulness and logical structure of these statements?

a) Both statements are true, as they accurately describe properties of real numbers involving square.
b) Statement 1 is true because it correctly describe the existence of a square for every real number. Statement 2 is false because it is incorrectly suggests a single square value for all real number.
c) Statement 1 is false because it misuses the existential quantifier statement 2 is true because it correctly identifies a single value of y applicable to all x.
d) Both statements are false due to misunderstanding of the relationship between the quantifiers and the mathematical operations involved.

57. Suppose you are given a task of proving that for all natural numbers n_1 the sum of the first n natural numbers is given by the formula $\frac{n(n+1)}{2}$. Which principle would you primarily use to prove this statement and what is the first step in that proof?

a) Use the rule of product, starting by demonstrating the formula works for the two natural numbers.
b) Apply mathematical induction, beginning with verifying the formula for the base case n = 1
c) Employ the rule of sum, starting with the assumption that the formula holds in the first n – 1 natural numbers.
d) Utilize the Binomial theorem, beginning with expanding the binomial $(x+1)^n$ for the base case n = 1.

58. If $f : A \rightarrow B$ and $g : B \rightarrow C$ are two functions such that f is onto, g is one-to-one, and the cardinality of set A is greater than the cardinality of set C. Which statement best describe the composition $gof : A \rightarrow C$?
a) The composition 'gof' is necessarily one-to-one
b) The composition 'gof' is necessarily onto
c) It is impossible for 'gof' to be both one-to-one and onto given the cardinality condition
d) The pigeon-hole principle guarantee that 'gof' must replicate some elements from A in C, making it not one-to-one

59. A deck of card is shuffled and laid out in a row using the principles of derangements, what is the probability that exactly two cards will be in their original position (assuming the original position is defined by the order of the deck before shuffling and each card has a unique position from 1 to 52)?

a) $\frac{!50}{52!}$ b) $\frac{!52}{52!}$ c) $\frac{52!-!52}{52!}$ d) $\frac{\binom{52}{2}\cdot !50}{52!}$

60. Given a complete graph k_n with n vertices where $n > 4$ if two vertices are removed along with all edges incident to them, which of the following statement best describes the resulting graphs properties?
a) The resulting graph remains a complete group
b) The resulting graph is guaranteed to be isomorphic to k_{n-2}
c) The resulting graph will have exactly two connected graph
d) The resulting graph can still be connected to but is no longer a complete graph.

61. Which of the following is not a non functional requirement?
a) Portability b) Security c) Scalability d) User Interaction

62. The process to gather the software requirements from client, analyse and document is known as
a) Software Engineering Process b) User Engineering Process
c) Requirement Elicitation Process d) Requirement Engineering Process

63. Which one of the following is not one of the RUP Life cycle phases?
a) Inception phase b) Iteration phase c) Elaboration phase d) Transition phase

64. How many diagrams are there in unified modelling language?
a) Six b) Seven c) Eight d) Nine

65. Agile Software Development is based on
a) Incremental Development b) Iterative Development
c) Linear Development d) Both Incremental and Iterative Development

66. Register R5 is used in a program to point to the top of a stack. The instruction to remove the top ten items from the stack is ______
a) Rem #10, SP b) Add #40, R5 c) Del #40, R5 d) Pop –R5, #10

67. In program – controlled I/O, synchronization between the processor and the I/O device is achieved by
a) Processor polling the device b) Device raising interrupt using IRQ line
c) Program raising interrupt using IRQ line d) Device sending Bus request signal

68. Which of the following main memory needs periodic refreshing?
a) EPROM b) SRAM c) DRAM d) PROM

69. In an n-bit ripple-carry adder sum out of the MSB bit is obtained after ______ gate delay.
a) n b) 2n c) 2n + 1 d) n – 1

70. Assume that in a typical single bus organization the propagation delays along the bus and through the ALU are 0.5 ns and 3 ns respectively. The setup time for the registers in 0.4 ns and the hold time is 0. What is the minimum clock period required?
a) 3.0 ns b) 2.1 ns c) 3.1 ns d) 3.9 ns

71. Which one of the following is the correct orders of growth?
a) $\log_2 n, n^2, n\log_2 n, n^3$ b) $n, \log_2 n, n\log_2 n, n^2$
c) $n^2, n^3, n\log_2 n, \log_2 n$ d) $\log_2 n, n\log_2 n, n^2, n^3$

72. The worst case complexity of Quick Sort Algorithm is
a) $\theta(n^3)$ b) $\theta(n\log_2 n)$ c) $\theta(n^2)$ d) $\theta(2^n)$

73. The level order of the 'heap' (maximum heap) constructed for the input 20, 40, 15, 30, 60 is
a) 40, 15, 60, 20, 30 b) 60, 40, 15, 20, 30
c) 60, 20, 15, 30, 40 d) 60, 40, 30, 20, 15

74. Which algorithm is used to find the Transitive closure of a given graph?
a) Floyd's Algorithm b) Depth First Search Algorithm
c) Kruskal's Algorithm d) Warshall's Algorithm

75. The type of tree used to solve the given problem by applying Backtracking Technique is called
a) Huffman Tree b) Binary Search Tree c) State Space Tree d) B^+ tree

76. When we restarting an operating system, it will
a) Terminates all the running programs completely
b) Restarts all the processes
c) Shuts down the operating system completely
d) All of the above

77. The operating system services are accessed by the interface provided by _______
a) System calls b) Library c) API d) Assembly instructions

78. Job control language statements are used to
a) Read the input from slow speed card reader to the high speed magnetic disk
b) Allocate the CPU to a job
c) Specify to the operating system, the beginning and end of a job in a batch
d) All of the above

79. IDL stands for
a) Interface Data Library b) Interface Direct Language
c) Interface Database Language d) Interface Definition Language

80. The threads of the operating system classify into
a) Kernel and User level b) Mainframe and Motherboard level
c) Security and Memory level d) OS and CPU level

81. The technique of temporarily delaying outgoing acknowledgements so that they can be hooked into the next outgoing data frame is called

a) Piggy backing b) Cyclic Redundancy check
c) Fletcher's checksum d) Parity check

82. A network with a bandwidth of 10 Mbps can pass only an average of 12,000 frames per minutes with each frame carrying an average of 10,000 bits. What is the throughput of this network,

a) 1 Mbps b) 2 Mbps c) 10 Mbps d) 12 Mbps

83. The minimum and maximum size of payload field in ethernet frame is

a) 46 bytes, 1000 bytes b) 64 bytes, 1500 bytes
c) 46 bytes, 1500 bytes d) 64 bytes, 1000 bytes

84. In RSA algorithm if p = 7, q = 11 and e = 13 then what will be the value of d?

a) 40 b) 13 c) 37 d) 23

85. Which of the following is not correct in relation to multi-destination routing?

a) is same as broadcast routing b) contains the list of all destinations
c) data is not sent by packets d) there are multiple receivers

86. ________ is a procedure or transformation that an object performs or is subject to

a) Inheritance b) Method c) Abstraction d) Operation

87. ________ is a description of the real world objects reflected within the system.

a) Domain model b) Application model c) Usecase model d) Activity model

88. _______ is the selective examination of certain aspects of a problem.

a) Inheritance b) Abstraction c) Polymorphism d) Encapsulation

89. Description of a group of links with common structure and common semantics is called _____

a) Links b) Association c) Model d) Reference

90. During ________ the developer makes strategic decisions with broad consequences.

a) Analysis b) System design c) Class Design d) None of these

91. Which level of locking provides the highest degree of concurrency in a relational database?

a) Page b) Table c) Row
d) All of the above allow same degree of concurrency.

92. What is the candidate key for the relation R with attributes A, B, C, D given the following functional dependencies: $AB \rightarrow C$ and $C \rightarrow D$

a) AB b) AC c) BC d) ABC

93. Consider the relation Auditorium (Name, Address, Capacity). Complete the following SQL which should always find the address of the auditorium with maximum capacity.

SELECT A1.ADDRESS FROM AUDITORIUM A1

a) where A1.Capacity >= ALL(select A2.Capacity from Auditorium A2)
b) where A1.Capacity >= ANY(select A2.Capacity from Auditorium A2)
c) where A1.Capacity >= ALL(select max(A2.Capacity) from Auditorium A2)
d) where A1.Capacity >= ANY(select max(A2.Capacity) from Auditorium A2)

94. Consider the relation R(A, B, C, D) and the set of functional dependencies:

$A \rightarrow B$; $B \rightarrow C$; $C \rightarrow D$; $D \rightarrow A$

Which of the following statements is TRUE regarding the normalization of relation R?

a) R is in 2NF but not 3NF
b) R is in 3NF but not BCNF
c) R is in BCNF but not 3NF
d) R is not in either 3NF or BCNF

95. Given the relations SUDENT(USN, Name, Sem, courseCode, CGPA) and COURSE(courseCode, courseName, courseURL), which of the following queries cannot be expressed using the basic relational algebra operations (σ, π, $\times$, $-$, U, P)

a) Course code of every student to which he/she has enrolled
b) Students whose name is not same as their course name.
c) Student with the highest CGPA
d) All students who have enrolled for a particular course code

96. Arrival of a customer to a queuing system is considered as

a) Entity
b) Event
c) Activity
d) Attribute

97. 40% of the assembled ink-jet printers are rejected at the inspection station. Find the probability that the first acceptable ink-jet is the third one inspected considering each inspection as a Bernoulli trial.

a) 0.1
b) 0.036
c) 0.081
d) 0.096

98. To check the uniformity of random numbers which of the following tests are used?

a) Kolmogorov – Smirnov Test
b) Chi-square Test
c) Both a and b
d) None of these

99. Assuming that the following data is from an exponential distribution, estimate the parameter 79, 3, 0.06, 0.12, 34, 5

a) 5
b) 20.2
c) 0.02
d) 0.0495

100. Interactive run controller is used for

a) Calibration of a model
b) Verification of a model
c) Validation of a model
d) None of these

* * * * *

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b) Sampling Design, Process Data, Identify Research Problem, Formulate Hypothesis
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b) Developing the Research Hypothesis
c) Formulation of research problem
d) Execution of the Project

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a) Convenience or accidental sampling
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c) Quota sampling
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37. Which of the following is not a data collection method?
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c) Interview d) Research question

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a) Geographical area b) Social life
c) Cultural group d) People or culture

39. Ogives is also known as?
a) Cumulative frequency curves b) Frequency distributions
c) Transcendental Curve d) All of these

40. Which of the following strategies is used to determine how common, widespread, or pervasive a certain trait, phenomenon, attitude, or opinion is in a demographic population?
a) Cross-sectional descriptive survey b) Factorial survey method
c) Total survey design d) Pooled surveys

41. In the process of conducting research 'Formulation of Hypothesis' is followed by
a) Statement of Objectives b) Analysis of Data
c) Selection of Research Tools d) Collection of Data

42. The data of research is _______
a) Qualitative only b) Quantitative only c) Neither of them d) Both a and b

43. _______- test is used to prove hypothesis of smaller sample.
a) t b) f c) z d) P

44. _____ tailed test is used when the researcher's interest is primarily on one side of the issue.
a) 1 b) 2 c) 3 d) 4

45. ______ is a part of the universe that can be used as respondents to survey.
a) Frame b) Sample c) Hypothesis d) Population

46. _______ Hypothesis states that there is no relationship between two or more variables
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47. _______ research helps to solve practical problems.
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48. The final stage in a survey is
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49. ______ mean refers to the value obtained by dividing the sum of the values of all items by the total
a) number of items b) Arithmetic c) Geometric d) Harmonic

50 _______ is used to analyse differences between group means and the associated procedures
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PART – II
Discipline Oriented Section
(Computer Science and Engineering / Information Science and Engineering)

[50 MARKS]

51. Which level of locking provides the highest degree of concurrency in a relational database?
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d) All of the above allow same degree of concurrency.

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$A \rightarrow B$; $B \rightarrow C$; $C \rightarrow D$; $D \rightarrow A$
Which of the following statements is TRUE regarding the normalization of relation R?
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63. The minimum and maximum size of payload field in ethernet frame is
a) 46 bytes, 1000 bytes b) 64 bytes, 1500 bytes
c) 46 bytes, 1500 bytes d) 64 bytes, 1000 bytes

64. In RSA algorithm if $p = 7$, $q = 11$ and $e = 13$ then what will be the value of d?
a) 40 b) 13 c) 37 d) 23

65. Which of the following is not correct in relation to multi-destination routing?
a) is same as broadcast routing b) contains the list of all destinations
c) data is not sent by packets d) there are multiple receivers

66. ________ is a procedure or transformation that an object performs or is subject to
a) Inheritance b) Method c) Abstraction d) Operation

67. ________ is a description of the real world objects reflected within the system.
a) Domain model b) Application model c) Usecase model d) Activity model

68. _______ is the selective examination of certain aspects of a problem.
a) Inheritance b) Abstraction c) Polymorphism d) Encapsulation

69. Description of a group of links with common structure and common semantics is called _____
a) Links b) Association c) Model d) Reference

70. During ________ the developer makes strategic decisions with broad consequences.
a) Analysis b) System design c) Class Design d) None of these

71. Which of the following is not a non functional requirement?
a) Portability b) Security c) Scalability d) User Interaction

72. The process to gather the software requirements from client, analyse and document is known as
a) Software Engineering Process b) User Engineering Process
c) Requirement Elicitation Process d) Requirement Engineering Process

73. Which one of the following is not one of the RUP Life cycle phases?
a) Inception phase b) Iteration phase c) Elaboration phase d) Transition phase

74. How many diagrams are there in unified modelling language?
a) Six b) Seven c) Eight d) Nine

75. Agile Software Development is based on
a) Incremental Development b) Iterative Development
c) Linear Development d) Both Incremental and Iterative Development

76. Register R5 is used in a program to point to the top of a stack. The instruction to remove the top ten items from the stack is ______
a) Rem #10, SP b) Add #40, R5 c) Del #40, R5 d) Pop –R5, #10

77. In program – controlled I/O, synchronization between the processor and the I/O device is achieved by
a) Processor polling the device b) Device raising interrupt using IRQ line
c) Program raising interrupt using IRQ line d) Device sending Bus request signal

78. Which of the following main memory needs periodic refreshing?
a) EPROM b) SRAM c) DRAM d) PROM

79. In an n-bit ripple-carry adder sum out of the MSB bit is obtained after ______ gate delay.
a) n b) 2n c) 2n + 1 d) n – 1

80. Assume that in a typical single bus organization the propagation delays along the bus and through the ALU are 0.5 ns and 3 ns respectively. The setup time for the registers in 0.4 ns and the hold time is 0. What is the minimum clock period required?
a) 3.0 ns b) 2.1 ns c) 3.1 ns d) 3.9 ns

81. Consider a single dimensional array A[1..1000]. Base address 2000 and the size of each element is 4. Find the location of A[50]
a) 2186 b) 2126 c) 2096 d) 2196

82. Which data structure is used to remove recursion from a program?
a) Array b) Stack c) Queue d) List

83. Which of the following data structure is used to implement a priority queue in an efficient way?
a) Binary Heap b) Array c) Dequeue d) None of these

84. A linked list has O(1) time complexity for the following operation.
a) Insertion at the middle b) Deletion at the end
c) Insertion at the beginning d) None of these

85. What is the average time complexity of merge sort?
a) $\theta(\log n)$ b) $\theta(n \log n)$ c) $\theta(n^2)$ d) None of these

86. In mathematical logic clues a professor discuss the relationship between statements involving universal and existential quantifier consider following 2 statements:

1) "For every real number x, there exists a real number y such that $y = x^2$".
2) "There exists a real number y such that for every real number x, $y = x^2$".

Which of the following best evaluate the truthfulness and logical structure of these statements?

a) Both statements are true, as they accurately describe properties of real numbers involving square.

b) Statement 1 is true because it correctly describe the existence of a square for every real number. Statement 2 is false because it is incorrectly suggests a single square value for all real number.

c) Statement 1 is false because it misuses the existential quantifier statement 2 is true because it correctly identifies a single value of y applicable to all x.

d) Both statements are false due to misunderstanding of the relationship between the quantifiers and the mathematical operations involved.

87. Suppose you are given a task of proving that for all natural numbers n_1 the sum of the first n natural numbers is given by the formula $\frac{n(n+1)}{2}$. Which principle would you primarily use to prove this statement and what is the first step in that proof?

a) Use the rule of product, starting by demonstrating the formula works for the two natural numbers.

b) Apply mathematical induction, beginning with verifying the formula for the base case $n = 1$

c) Employ the rule of sum, starting with the assumption that the formula holds in the first $n - 1$ natural numbers.

d) Utilize the Binomial theorem, beginning with expanding the binomial $(x+1)^n$ for the base case $n = 1$.

88. If $f : A \rightarrow B$ and $g : B \rightarrow C$ are two functions such that f is onto, g is one-to-one, and the cardinality of set A is greater than the cardinality of set C. Which statement best describe the composition $gof : A \rightarrow C$?

a) The composition 'gof' is necessarily one-to-one

b) The composition 'gof' is necessarily onto

c) It is impossible for 'gof' to be both one-to-one and onto given the cardinality condition

d) The pigeon-hole principle guarantee that 'gof' must replicate some elements from A in C, making it not one-to-one

89. A deck of card is shuffled and laid out in a row using the principles of derangements, what is the probability that exactly two cards will be in their original position (assuming the original position is defined by the order of the deck before shuffling and each card has a unique position from 1 to 52)?

a) $\frac{!50}{52!}$ b) $\frac{!52}{52!}$ c) $\frac{52!-!52}{52!}$ d) $\frac{\binom{52}{2}\cdot !50}{52!}$

90. Given a complete graph k_n with n vertices where $n > 4$ if two vertices are removed along with all edges incident to them, which of the following statement best describes the resulting graphs properties?

a) The resulting graph remains a complete group

b) The resulting graph is guaranteed to be isomorphic to k_{n-2}

c) The resulting graph will have exactly two connected graph

d) The resulting graph can still be connected to but is no longer a complete graph.

91. Which one of the following is the correct orders of growth?

a) $\log_2 n$, n^2, $n \log_2 n$, n^3
b) n, $\log_2 n$, $n \log_2 n$, n^2
c) n^2, n^3, $n \log_2 n$, $\log_2 n$
d) $\log_2 n$, $n \log_2 n$, n^2, n^3

92. The worst case complexity of Quick Sort Algorithm is

a) $\theta(n^3)$
b) $\theta(n \log_2 n)$
c) $\theta(n^2)$
d) $\theta(2^n)$

93. The level order of the 'heap' (maximum heap) constructed for the input 20, 40, 15, 30, 60 is

a) 40, 15, 60, 20, 30
b) 60, 40, 15, 20, 30
c) 60, 20, 15, 30, 40
d) 60, 40, 30, 20, 15

94. Which algorithm is used to find the Transitive closure of a given graph?

a) Floyd's Algorithm
b) Depth First Search Algorithm
c) Kruskal's Algorithm
d) Warshall's Algorithm

95. The type of tree used to solve the given problem by applying Backtracking Technique is called

a) Huffman Tree
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96. When we restarting an operating system, it will

a) Terminates all the running programs completely
b) Restarts all the processes
c) Shuts down the operating system completely
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97. The operating system services are accessed by the interface provided by _______

a) System calls
b) Library
c) API
d) Assembly instructions

98. Job control language statements are used to

a) Read the input from slow speed card reader to the high speed magnetic disk
b) Allocate the CPU to a job
c) Specify to the operating system, the beginning and end of a job in a batch
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a) Interface Data Library
b) Interface Direct Language
c) Interface Database Language
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100. The threads of the operating system classify into

a) Kernel and User level
b) Mainframe and Motherboard level
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* * * * *

VTU-ETR Seat No.

Question Paper Version : C

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) May – 2024

DO NOT USE MOBILE PHONES

Computer Science and Engineering / Information Science and Engineering

Time: 3 hrs. Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
10. Partial or incomplete shading/darkening of the bubble be avoided.
11. Use of whiteners/correction fluid to change the shaded/filled checkbox bubble to fill another is prohibited.
12. Erasing of filled checkbox bubble/s to fill another is prohibited.
13. Any type of writing, scribbling etc., on the question paper and ticking the response of a question/s are prohibited. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
14. Only non-programmable scientific calculator is permitted. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

PART – I

(Research Methodology)

[50 Marks]

1. Z test is used to test hypothesis when the sample size is ______
a) more than 30 b) less than 30
c) less than 50 d) none of these

2. The mode of 2, 5, 6, 4, 2, 7, 2, 5 is
a) 5 b) 6 c) 2 d) 7

3. Hypothesis should be ______
a) Any statement b) Empirically testable
c) Not empirically testable d) Can't say

4. What is the purpose of doing research?
a) To identify problem b) To find the solution
c) Both a and b d) None of these

5. Which ONE of these sampling methods is a probability method?
a) Quota b) Judgment
c) Convenience d) Simple random

6. _______ is the first step of the Research process
a) Formulation of a problem b) Collection of Data
c) Editing and Coding d) Selection of a problem

7. If the critical region is located equally in both sides of the sampling distribution of test statistics, the test is called
a) One-tailed b) Two-tailed c) Right tailed d) Left tailed

8. The standard deviation is always ______
a) Negative b) Positive c) Sometimes negative d) None of these

9. Mean, Median and Mode are
a) Measures of deviation b) Measures of dispersion
c) Measures of control tendency d) Measures of central tendency

10. A tentative proposition subject to test is
a) Variable b) Hypothesis c) Data d) Concept

11. Which of the following research is also known as collaborative research?
a) Analytical research b) Applied research
c) Action research d) Descriptive research

12. Cross-sectional surveys are divided into two types, which are?
a) Exploratory and causal b) Functional and Dysfunctional
c) Relational and logical d) Descriptive and analytical

13. What are the main purposes of data analysis?
a) Description b) Construction of Measurement Scale
c) Generating empirical relationships d) Explanation and prediction

14. __________ involve a set of predetermined questions and highly standardized techniques of recording?
a) Structured interview b) Unstructured interview
c) Interview guided d) All of these

15. Which of the following is a primary step in social science research?
a) Preparing the Research Design b) Developing the Research Hypothesis
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16. Find an example of probability sampling.
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21. A Hypothesis that develops while planning the research is
a) Null Hypothesis b) Working Hypothesis
c) Relational Hypothesis d) Descriptive Hypothesis

22. The fullform of APA is ______
a) American Physical Association b) American Psychological Association
c) American Psychological Amendment d) American Psychological Agreement

23. Questions in which only two alternatives are possible is called
a) Dichotomous questions b) Open-ended questions
c) Structured questions d) MCQs

24. Questionnaire is a
a) Research method b) Measurement technique
c) Tool for data collection d) Data analysis technique

25. Which of the following order is recommended in the flowchart of the research process?
a) Formulate Hypothesis, Sampling Design, Process Data, Identify Research Problem
b) Sampling Design, Process Data, Identify Research Problem, Formulate Hypothesis
c) Formulate Hypothesis, Process Data, Identify Research Problem, Sampling Design
d) Identify Research Problem, Formulate Hypothesis, Sampling Design, Process Data

26. If $0 < r < 1$ means there is ______
a) Negative correlation b) Positive correlation
c) No correlation d) All of these

27. _______ is a process of checking errors and omissions in data collection.
a) Editing b) Coding c) Tabulation d) None of these

28. _______ is the difference between highest and lowest value in a series of data.
a) Median b) Range c) Mode d) None of these

29. If Ho is true and we reject it is called
a) Type-I error b) Type-II error c) Standard error d) None of these

30. Sending a Questionnaire to a respondent with a request to complete and return by post is called
a) Interview b) Observation c) A Mail Survey d) Panel

31. In the process of conducting research 'Formulation of Hypothesis' is followed by
a) Statement of Objectives b) Analysis of Data
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40. _______ is used to analyse differences between group means and the associated procedures
a) ANOVA b) Chi square c) Z-test d) correlation

41. Quantitative research involves conducting research using the following :
a) Feelings and hidden emotions, expectations
b) Facts, Numbers and numerical data
c) Humour and gratitude
d) Ethics, Accuracy and discipline

42. Which of the following best describes quantitative research?
a) the collection of non-numerical data
b) an attempt to confirm the researcher's hypotheses
c) exploratory research
d) research that attempts to generate a new theory

43. Which of the following includes examples of quantitative variables?
a) age, temperature, income, height
b) grade point average, anxiety level, reading performance
c) gender, religion, ethnic group
d) both a and b

44. Quantitative research only works if
a) You talk to the right number of people
b) You talk to the right type of people
c) You ask the right question and analyse the data you get in the right way
d) All of the above

45. An image, perception or concept that is capable of measurement is called _______
a) Scale
b) Hypothesis
c) Type
d) Variable

46. Why do you need to review the existing literature?
a) To make sure you have a long list of references
b) Because without it, you could never reach the required word-count
c) To help in your general studying
d) To find out what is already known about your area of interest

47. To pursue the research, which of the following is priorly required?
a) Developing a research design
b) Formulating a research question
c) Deciding about the data analysis procedure
d) Formulating a research hypothesis

48. A statement of the quantitative research question should:
a) Extend the statement of purpose by specifying exactly the question (the researcher will address)
b) Help the research in selecting appropriate participants, research methods, measures and materials
c) Specify the variables of interest
d) None of the above

49. Concept is of two types
a) Abstract and coherent
b) Concrete and coherent
c) Abstract and concrete
d) None of these

50 _________ is concerned with discovering and testing certain variables with respect to their association or disassociation.
a) Exploratory
b) Descriptive
c) Diagnostic
d) Descriptive and diagnostic

PART – II

Discipline Oriented Section

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a) Course code of every student to which he/she has enrolled
b) Students whose name is not same as their course name.
c) Student with the highest CGPA
d) All students who have enrolled for a particular course code

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a) The composition 'gof' is necessarily one-to-one
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* * * * *

VTU-ETR Seat No. | | | | | | |

Question Paper Version : D

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) May – 2024

DO NOT USE MOBILE PHONES

Computer Science and Engineering / Information Science and Engineering

Time: 3 hrs. Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
10. Partial or incomplete shading/darkening of the bubble be avoided.
11. Use of whiteners/correction fluid to change the shaded/filled checkbox bubble to fill another is prohibited.
12. Erasing of filled checkbox bubble/s to fill another is prohibited.
13. Any type of writing, scribbling etc., on the question paper and ticking the response of a question/s are prohibited. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
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15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

PART – I

(Research Methodology)

[50 Marks]

1. In the process of conducting research 'Formulation of Hypothesis' is followed by
a) Statement of Objectives b) Analysis of Data
c) Selection of Research Tools d) Collection of Data

2. The data of research is _______
a) Qualitative only b) Quantitative only c) Neither of them d) Both a and b

3. _______- test is used to prove hypothesis of smaller sample.
a) t b) f c) z d) P

4. _____ tailed test is used when the researcher's interest is primarily on one side of the issue.
a) 1 b) 2 c) 3 d) 4

5. ______ is a part of the universe that can be used as respondents to survey.
a) Frame b) Sample c) Hypothesis d) Population

6. _______ Hypothesis states that there is no relationship between two or more variables
a) Null b) Alternative c) Negative Page d) Positive

7. _______ research helps to solve practical problems.
a) Qualitative b) Applied c) Basic d) Descriptive

8. The final stage in a survey is
a) Field work b) Assignment c) Calculation d) Reporting

9. ______ mean refers to the value obtained by dividing the sum of the values of all items by the total
a) number of items b) Arithmetic c) Geometric d) Harmonic

10. _______ is used to analyse differences between group means and the associated procedures
a) ANOVA b) Chi square c) Z-test d) correlation

11. A Hypothesis that develops while planning the research is
a) Null Hypothesis b) Working Hypothesis
c) Relational Hypothesis d) Descriptive Hypothesis

12. The fullform of APA is ______
a) American Physical Association b) American Psychological Association
c) American Psychological Amendment d) American Psychological Agreement

13. Questions in which only two alternatives are possible is called
a) Dichotomous questions b) Open-ended questions
c) Structured questions d) MCQs

14. Questionnaire is a
a) Research method b) Measurement technique
c) Tool for data collection d) Data analysis technique

15. Which of the following order is recommended in the flowchart of the research process?
a) Formulate Hypothesis, Sampling Design, Process Data, Identify Research Problem
b) Sampling Design, Process Data, Identify Research Problem, Formulate Hypothesis
c) Formulate Hypothesis, Process Data, Identify Research Problem, Sampling Design
d) Identify Research Problem, Formulate Hypothesis, Sampling Design, Process Data

16. If $0 < r < 1$ means there is ______
a) Negative correlation b) Positive correlation
c) No correlation d) All of these

17. _______ is a process of checking errors and omissions in data collection.
a) Editing b) Coding c) Tabulation d) None of these

18. _______ is the difference between highest and lowest value in a series of data.
a) Median b) Range c) Mode d) None of these

19. If Ho is true and we reject it is called
a) Type-I error b) Type-II error c) Standard error d) None of these

20. Sending a Questionnaire to a respondent with a request to complete and return by post is called
a) Interview b) Observation c) A Mail Survey d) Panel

21. Z test is used to test hypothesis when the sample size is ______
a) more than 30 b) less than 30 c) less than 50 d) none of these

22. The mode of 2, 5, 6, 4, 2, 7, 2, 5 is
a) 5 b) 6 c) 2 d) 7

23. Hypothesis should be ______
a) Any statement b) Empirically testable
c) Not empirically testable d) Can't say

24. What is the purpose of doing research?
a) To identify problem b) To find the solution
c) Both a and b d) None of these

25. Which ONE of these sampling methods is a probability method?
a) Quota b) Judgment
c) Convenience d) Simple random

26. _______ is the first step of the Research process
a) Formulation of a problem b) Collection of Data
c) Editing and Coding d) Selection of a problem

27. If the critical region is located equally in both sides of the sampling distribution of test statistics, the test is called
a) One-tailed b) Two-tailed c) Right tailed d) Left tailed

28. The standard deviation is always ______
a) Negative b) Positive c) Sometimes negative d) None of these

29. Mean, Median and Mode are
a) Measures of deviation
b) Measures of dispersion
c) Measures of control tendency
d) Measures of central tendency

30. A tentative proposition subject to test is
a) Variable b) Hypothesis c) Data d) Concept

31. Quantitative research involves conducting research using the following :
a) Feelings and hidden emotions, expectations
b) Facts, Numbers and numerical data
c) Humour and gratitude
d) Ethics, Accuracy and discipline

32. Which of the following best describes quantitative research?
a) the collection of non-numerical data
b) an attempt to confirm the researcher's hypotheses
c) exploratory research
d) research that attempts to generate a new theory

33. Which of the following includes examples of quantitative variables?
a) age, temperature, income, height
b) grade point average, anxiety level, reading performance
c) gender, religion, ethnic group
d) both a and b

34. Quantitative research only works if
a) You talk to the right number of people
b) You talk to the right type of people
c) You ask the right question and analyse the data you get in the right way
d) All of the above

35. An image, perception or concept that is capable of measurement is called _______
a) Scale b) Hypothesis c) Type d) Variable

36. Why do you need to review the existing literature?
a) To make sure you have a long list of references
b) Because without it, you could never reach the required word-count
c) To help in your general studying
d) To find out what is already known about your area of interest

37. To pursue the research, which of the following is priorly required?
a) Developing a research design
b) Formulating a research question
c) Deciding about the data analysis procedure
d) Formulating a research hypothesis

38. A statement of the quantitative research question should:
a) Extend the statement of purpose by specifying exactly the question (the researcher will address)
b) Help the research in selecting appropriate participants, research methods, measures and materials
c) Specify the variables of interest
d) None of the above

39. Concept is of two types

a) Abstract and coherent
b) Concrete and coherent
c) Abstract and concrete
d) None of these

40. _________ is concerned with discovering and testing certain variables with respect to their association or disassociation.

a) Exploratory
b) Descriptive
c) Diagnostic
d) Descriptive and diagnostic

41. Which of the following research is also known as collaborative research?

a) Analytical research
b) Applied research
c) Action research
d) Descriptive research

42. Cross-sectional surveys are divided into two types, which are?

a) Exploratory and causal
b) Functional and Dysfunctional
c) Relational and logical
d) Descriptive and analytical

43. What are the main purposes of data analysis?

a) Description
b) Construction of Measurement Scale
c) Generating empirical relationships
d) Explanation and prediction

44. __________ involve a set of predetermined questions and highly standardized techniques of recording?

a) Structured interview
b) Unstructured interview
c) Interview guided
d) All of these

45. Which of the following is a primary step in social science research?

a) Preparing the Research Design
b) Developing the Research Hypothesis
c) Formulation of research problem
d) Execution of the Project

46. Find an example of probability sampling.

a) Convenience or accidental sampling
b) Purposive or judgmental sampling
c) Quota sampling
d) Stratified random sampling

47. Which of the following is not a data collection method?

a) Observation
b) Schedules
c) Interview
d) Research question

48. The term ‘ethno’ refers to

a) Geographical area
b) Social life
c) Cultural group
d) People or culture

49. Ogives is also known as?

a) Cumulative frequency curves
b) Frequency distributions
c) Transcendental Curve
d) All of these

50 Which of the following strategies is used to determine how common, widespread, or pervasive a certain trait, phenomenon, attitude, or opinion is in a demographic population?

a) Cross-sectional descriptive survey
b) Factorial survey method
c) Total survey design
d) Pooled surveys

PART – II
Discipline Oriented Section
(Computer Science and Engineering / Information Science and Engineering)

[50 MARKS]

51. The technique of temporarily delaying outgoing acknowledgements so that they can be hooked into the next outgoing data frame is called
a) Piggy backing b) Cyclic Redundancy check
c) Fletcher's checksum d) Parity check

52. A network with a bandwidth of 10 Mbps can pass only an average of 12,000 frames per minutes with each frame carrying an average of 10,000 bits. What is the throughput of this network,
a) 1 Mbps b) 2 Mbps c) 10 Mbps d) 12 Mbps

53. The minimum and maximum size of payload field in ethernet frame is
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71. Which level of locking provides the highest degree of concurrency in a relational database?
a) Page b) Table c) Row
d) All of the above allow same degree of concurrency.

72. What is the candidate key for the relation R with attributes A, B, C, D given the following functional dependencies: $AB \rightarrow C$ and $C \rightarrow D$
a) AB b) AC c) BC d) ABC

73. Consider the relation Auditorium (Name, Address, Capacity). Complete the following SQL which should always find the address of the auditorium with maximum capacity.
SELECT A1.ADDRESS FROM AUDITORIUM A1
a) where A1.Capacity >= ALL(select A2.Capacity from Auditorium A2)
b) where A1.Capacity >= ANY(select A2.Capacity from Auditorium A2)
c) where A1.Capacity >= ALL(select max(A2.Capacity) from Auditorium A2)
d) where A1.Capacity >= ANY(select max(A2.Capacity) from Auditorium A2)

74. Consider the relation R(A, B, C, D) and the set of functional dependencies:
$A \rightarrow B$; $B \rightarrow C$; $C \rightarrow D$; $D \rightarrow A$
Which of the following statements is TRUE regarding the normalization of relation R?
a) R is in 2NF but not 3NF b) R is in 3NF but not BCNF
c) R is in BCNF but not 3NF d) R is not in either 3NF or BCNF

75. Given the relations SUDENT(USN, Name, Sem, courseCode, CGPA) and COURSE(courseCode, courseName, courseURL), which of the following queries cannot be expressed using the basic relational algebra operations (σ, π, $\times$, $-$, U, P)
a) Course code of every student to which he/she has enrolled
b) Students whose name is not same as their course name.
c) Student with the highest CGPA
d) All students who have enrolled for a particular course code

76. Arrival of a customer to a queuing system is considered as
a) Entity b) Event c) Activity d) Attribute

77. 40% of the assembled ink-jet printers are rejected at the inspection station. Find the probability that the first acceptable ink-jet is the third one inspected considering each inspection as a Bernoulli trial.
a) 0.1 b) 0.036 c) 0.081 d) 0.096

78. To check the uniformity of random numbers which of the following tests are used?
a) Kolmogorov – Smirnov Test b) Chi-square Test
c) Both a and b d) None of these

79. Assuming that the following data is from an exponential distribution, estimate the parameter
79, 3, 0.06, 0.12, 34, 5
a) 5 b) 20.2 c) 0.02 d) 0.0495

80. Interactive run controller is used for
a) Calibration of a model b) Verification of a model
c) Validation of a model d) None of these

81. Which one of the following is the correct orders of growth?
a) $\log_2 n$, n^2, $n\log_2 n$, n^3 b) n, $\log_2 n$, $n\log_2 n$, n^2
c) n^2, n^3, $n\log_2 n$, $\log_2 n$ d) $\log_2 n$, $n\log_2 n$, n^2, n^3

82. The worst case complexity of Quick Sort Algorithm is
a) $\theta(n^3)$ b) $\theta(n\log_2 n)$ c) $\theta(n^2)$ d) $\theta(2^n)$

83. The level order of the 'heap' (maximum heap) constructed for the input 20, 40, 15, 30, 60 is
a) 40, 15, 60, 20, 30 b) 60, 40, 15, 20, 30
c) 60, 20, 15, 30, 40 d) 60, 40, 30, 20, 15

84. Which algorithm is used to find the Transitive closure of a given graph?
a) Floyd's Algorithm b) Depth First Search Algorithm
c) Kruskal's Algorithm d) Warshall's Algorithm

85. The type of tree used to solve the given problem by applying Backtracking Technique is called
a) Huffman Tree b) Binary Search Tree c) State Space Tree d) B^+ tree

86. When we restarting an operating system, it will
a) Terminates all the running programs completely
b) Restarts all the processes
c) Shuts down the operating system completely
d) All of the above

87. The operating system services are accessed by the interface provided by _______
a) System calls b) Library c) API d) Assembly instructions

88. Job control language statements are used to
a) Read the input from slow speed card reader to the high speed magnetic disk
b) Allocate the CPU to a job
c) Specify to the operating system, the beginning and end of a job in a batch
d) All of the above

89. IDL stands for
a) Interface Data Library
b) Interface Direct Language
c) Interface Database Language
d) Interface Definition Language

90. The threads of the operating system classify into
a) Kernel and User level
b) Mainframe and Motherboard level
c) Security and Memory level
d) OS and CPU level

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100. Assume that in a typical single bus organization the propagation delays along the bus and through the ALU are 0.5 ns and 3 ns respectively. The setup time for the registers in 0.4 ns and the hold time is 0. What is the minimum clock period required?
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